

1. hard process (ME): $q\bar{q} \rightarrow b\bar{b}$

run 1 lumi 1 event 1

2 particles drawn (marker size $\sim \log p_T$; label = p_T [GeV] for $p_T > 5$ GeV)

b quark (status 23): $p_T = 113.22$ GeV, $\eta = -1.44$, $\phi = -2.15$

$\bar{b}$ quark (status 23): $p_T = 113.22$ GeV, $\eta = -3.75$, $\phi = 0.99$

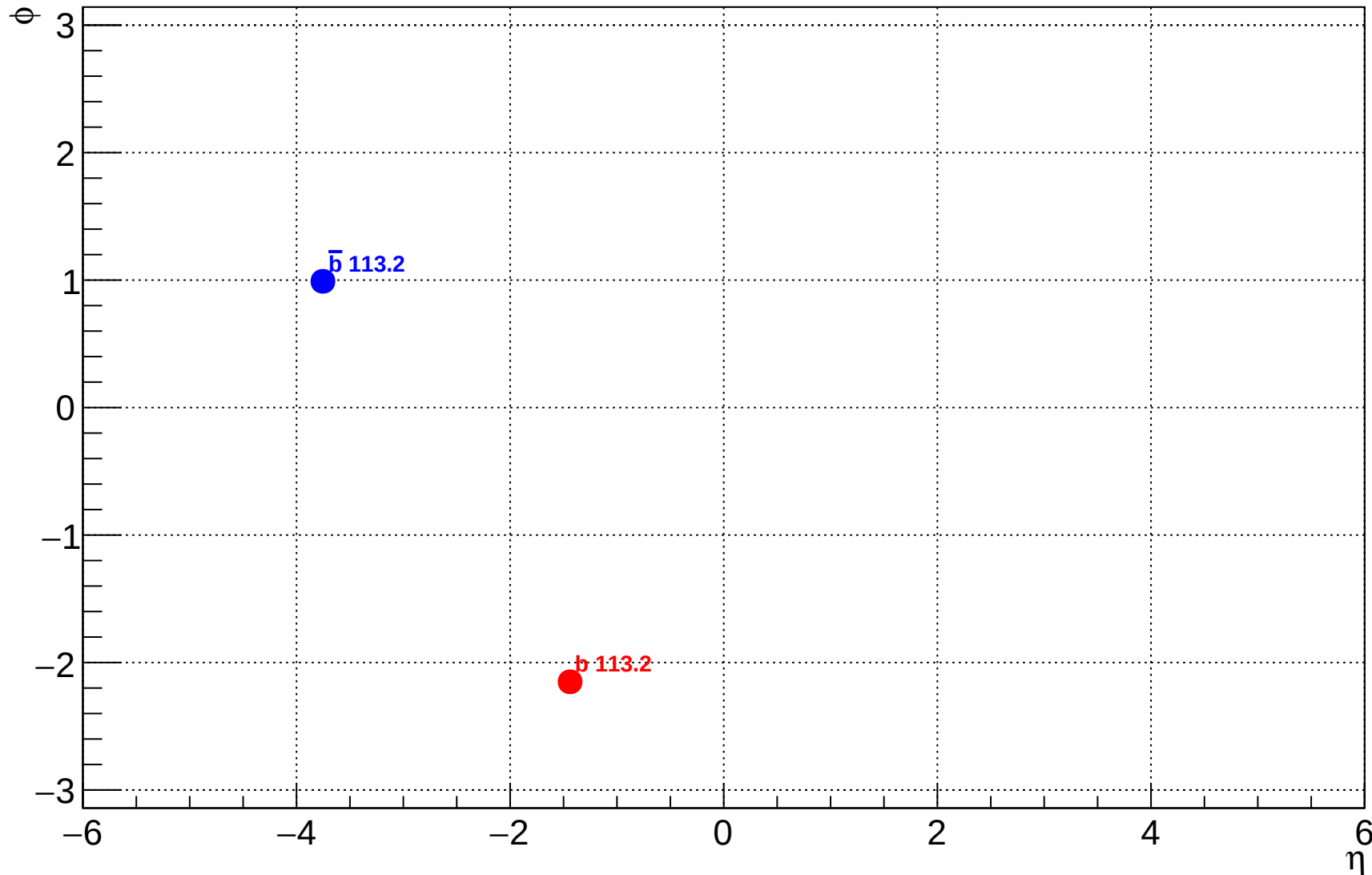
net $b+\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})| = 0.00$ GeV ($p_x = -0.00$, $p_y = 0.00$), $\Sigma|p_T| = 226.44$ GeV, $\Delta\phi(b,\bar{b}) = 3.14$

sample ME_PS_MPI

ISR on, FSR on, MPI on

Had off, Decay off

pTHatMin 100 GeV



particle origin (number drawn)

● b system (1)

● $\bar{b}$ system (1)

★ b + $\bar{b}$ (vector sum)

marker size:

● $p_T = 1$ GeV

● $p_T = 10$ GeV

● $p_T = 100$ GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays

3. + MPI / beam remnants

run 1 lumi 1 event 1

169 particles drawn (marker size $\sim \log p_T$; label = p_T [GeV] for $p_T > 5$ GeV), 19 with $|\eta| > 6$ drawn on the border (x)

b quark (status 52): $p_T = 91.59$ GeV, $\eta = -1.57$, $\phi = -2.08$

$\bar{b}$ quark (status 52): $p_T = 118.57$ GeV, $\eta = -3.73$, $\phi = 1.02$

net $b+\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})| = 27.29$ GeV ($p_x = 17.19$, $p_y = 21.19$), $\Sigma|p_T| = 210.16$ GeV, $\Delta\phi(b, \bar{b}) = 3.10$

sample ME_PS_MPI

ISR on, FSR on, MPI on

Had off, Decay off

pTHatMin 100 GeV

